NADUN DE SILVA

Team Lead and Software Engineer

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Highly motivated Software Engineer with 4+ years of experience in cloud native application development and data analytics, performing above the top 5% of the employees at WSO2. Experience in designing large scale reliable architecture, UX, developing and deploying cloud native applications in production environments. Great track record in delivering quality outcomes while balancing company goals and user requirements. Experience in leading 6 engineers.

Work Experience

Associate Technical Lead

WSO2 June 2021 – Present

- Achieved the Sustained Outstanding Contribution Award for the third consecutive year — currently awarded only for the top 5% employees in the company.
- Led another engineer, designed and developed the minimum viable features for Choreo online editor's automatic resource scheduling, within 1.5 months, using Kubernetes and GoLang.
- Reduced the startup time of the Kubernetes resources of the Choreo Editors by 80%, increasing the overall user experience.
- Reduced the MsSQL DB utilization by 60% by introducing a Redis cache and randomization of cache expiry times, increasing the number of users the system can handle.
- Led 3 to 6 engineers working on Choreo Observability and Choreo Editor scheduling, which had a proven track record as being some of the most stable areas in Choreo.

Senior Software Engineer

WSO2 July 2019 – June 2021

- Led 2 engineers, designed and developed the minimum viable features for Choreo Observability within 3 months, creating a scalable backbone for Choreo Observability.
- Designed the architecture of Choreo observability storages with automated archiving to reduce the cost for the company while storing enough data for Machine Learning use cases.
- Redesigned and rewrote the observability instrumentation at Ballerina compiler intermediate representation level (previously done at Java bytecode generation level), within 1 month, to collect additional information.

Software Engineer

WSO2 Jan 2018 – July 2019

- Delivered the Cellery Observability minimum viable features within 2 months for observing microservice composites, using Kubernetes, Istio, OpenTracing and Envoy.
- Led the implementation of Cellery developer tools.
- Designed Cellery Hub backed by a Docker Registry as storage.

Software Engineering Intern

WSO2 July 2016 – Dec 2016

- Implemented a Notebook prototype for data analytics.
- Implemented several Stream Processing extensions for Siddhi.

Technical Skills

- **Programming Languages** — Java, GoLang, Python, JavaScript, TypeScript
- **Knowledge Domains** — Cloud Computing, Time-series Analysis, Observability, Deep Learning, Anomaly Detection
- **Frameworks and Tools** — Git, Gradle, JMeter, Express, React, TensorFlow, Numpy, Pandas, Azure Event Hub (Kafka)
- **Storages** — Time-series Databases (Influx DB, Azure Data Explorer, Azure Time Series Insights), Data Lakes (Azure Data Lake), Relational Databases (MySQL, MsSQL), Redis
- **Deployment Tools and Technologies** — Kubernetes, Kustomize, Docker, GitOps, Prometheus, Jaeger

Licenses

Certified Kubernetes Administrator

The Linux Foundation Dec 2020

Certified Kubernetes Application Developer

The Linux Foundation Jan 2020

Education

B.Sc. (Hons.) in Engineering (Computer Science and Engineering)

University of Moratuwa 2014 – 2017

- Achieved a GPA of 3.85 out of 4.20 obtaining a First Class.
- Placements in Dean's List in 6 out of 8 semesters at University of Moratuwa.
- Carried out an academic research on Anomaly Detection using Deep Learning techniques (VAE and Bidirectional GAN) with 2 publications.
- Awarded Global Finalist (Galactic Impact) in NASA Space Apps Challenge 2017.
- Successfully completed Google Summer of Code 2017.

Certifications

Fundamentals of Reinforcement Learning
Amii, University of Alberta September 2021

Build Basic Generative Adversarial Networks (GANs)
DeepLearning.AI June 2021

Deep Learning Specialization
DeepLearning.AI June 2021

Projects

Choreo

Digital innovation platform that allows you to develop, deploy, and manage cloud native applications at scale. Choreo manages the complete lifecycle of an application and deploys them in a production grade environment along with automated observability.

Cellery

Framework for developing Code first observable, secure composites for Kubernetes. Cellery implemented cell based architecture to improve the developer experience and collaboration. It supported observing the composites and a drill down view for detecting and debugging issues easily.

Ballerina

An open-source programming language for cloud native application development with builtin support for automated compiler level Observability instrumentation.

K8s Replicator

A Kubernetes controller implementing a control loop which copies Kubernetes resources across namespaces. This allowed SREs to manage common resources which needed to be shared across multiple namespaces (such as TLS secrets), to be managed centrally. The controller was built to support extending to any Kubernetes resource easily.

Publications

- T. Kumarage, S. Ranathunga, C. Kuruppu, N. De Silva and M. Ranawaka. (2019). Generative Adversarial Networks (GAN) based Anomaly Detection in Industrial Software Systems. In *2019 Moratuwa Engineering Research Conference (MERCon)* (pp. 43–48). doi:10.1109/MERCon.2019.8818750
- T. Kumarage, N. De Silva, M. Ranawaka, C. Kuruppu and S. Ranathunga. (2018). Anomaly Detection in Industrial Software Systems - Using Variational Autoencoders. In *Proceedings of the 7th International Conference on Pattern Recognition Applications and Methods - Volume 1: ICPRAM* (pp. 440–447). INSTICC. doi:10.5220/0006600304400447

Soft Skills

Leadership Skills · Presentation Skills · Communication Skills · Time Management

Languages

Sinhala
English

Interests

Deep Learning · Music · Reading · Martial Arts

Referees

Available Upon Request